Fit10 — Product Requirements Document

10-Minute Daily Home Workout & Adaptive Personal Trainer

PRODUCTION BASELINE • VERSION 1.0.0

APP IDENTIFIER	TARGET OS / SDK	ARCHITECTURE	DATABASE / ENGINE
com.fit10.app	Android 14+ (Min SDK 24, Target 37)	MVVM + Clean Arch (Jetpack Compose)	Room SQLite + Reactive StateFlow
MONETIZATION	**CONTENT CATALOG**	**EXECUTION MODEL**	**DOCUMENT DATE**
Google Play Billing v9.1.0 (Pro Pass)	21 Exercises • 10 Workouts • 2 Programs	100% Offline-First (No Mandatory Cloud)	October 2026

1 Executive Summary & Product Vision

Fit10 addresses the primary bottleneck in home fitness adherence: *time friction and chronic burnout*. While conventional fitness applications prescribe rigid 45–60 minute sessions requiring equipment or gyms, Fit10 delivers high-density, scientifically sequenced **10-minute micro-workouts** performed entirely with bodyweight.

> **Core Value Proposition**
>
> "Ten minutes of scientifically programmed daily movement that adapts to your joint comfort, prevents overtraining through muscle recovery awareness, and builds an unbreakable exercise habit."

By integrating an automated deload cycle, muscle overlap protection (ensuring back-to-back workouts do not fatigue identical muscle groups), and real-time exercise substitution, Fit10 ensures every user makes steady progress without risk of overuse injury or demotivation.

2 Target Demographics & User Personas

Persona	Primary Motivation	Pain Points	Fit10 Solution
The Time-Constrained Professional	Maintain health and energy without sacrificing calendar blocks or commuting to gyms.	Unpredictable meetings, fatigue after long work hours, guilt over skipping 1-hour workouts.	Zero-setup 10-minute sessions with automated timers and customizable daily reminder schedules.
The Habit Starter / Beginner	Build a sustainable, non-intimidating daily fitness habit from home.	Severe post-workout soreness, joint discomfort, complex gym jargon, gym anxiety.	14-Day Kickstart guided track, clear form instructions, and joint-friendly low-impact regressions.
The Adaptive Athlete	Maintain functional conditioning, core stability, and posterior chain strength.	Plateaus caused by static routines; overtraining push/quad groups without pulling exercises.	Dynamic progression engine (+2 reps / +5s holds), balanced push/pull volume, and deload weeks.

3 Information Architecture & Navigation

Fit10 utilizes single-activity Android architecture driven by Jetpack Compose Navigation (`Fit10NavGraph`). The destination graph enforces seamless state restoration and deep linking:

- **Splash Screen:** Evaluates onboarding completion flag in DataStore; routes first-time users to Onboarding and returning users directly to Home.
- **Onboarding Flow:** 6-step guided assessment generating a personalized user profile and plan summary.
- **Home Dashboard:** Primary hub featuring daily recommended workout, streak rings, active program tracker, and focus area shortcuts.
- **Workout Preview & Detail:** Exercise breakdown with form cues, rep/duration targets, and one-tap exercise replacement.
- **Interval Workout Player:** High-precision ticker managing countdowns, work intervals, rest periods, audio tones, and haptic pulses.
- **Workout Summary & RPE Feedback:** Completion telemetry, calorie computation, perceived exertion input, and real-time engine adaptation messaging.
- **Discovery & Analytics:** Explore catalog, workout history calendar, reminder settings, profile, and subscription management.

4 Functional Requirements & Feature Specifications

4.1 Onboarding & Assessment Flow

A guided 6-step questionnaire implemented in `OnboardingViewModel` that captures the user's fitness profile and synthesizes their initial training program:

- **Step 1: Fitness Level:** Beginner, Intermediate, or Advanced baseline.
- **Step 2: Primary Fitness Goal:** Lose Weight, Build Strength, Improve Fitness, Flexibility, or Stay Active.
- **Step 3: Target Daily Duration:** 5, 10 (default), or 15 minutes.
- **Step 4: Available Equipment:** None / Bodyweight (default), Dumbbells, Resistance Bands, Yoga Mat.
- **Step 5: Exercise Exclusions / Physical Limitations:** Avoid jumping, avoid knee stress, avoid wrist loading, or none.
- **Step 6: Weekly Schedule:** Select active days per week (default 5 days) and recovery style (Active Recovery vs Complete Rest).

4.2 Intelligent Personalization & Adaptation Engine

The core algorithm (`PersonalizationEngine`) scores candidate workouts and recalibrates user volume using multi-factor heuristics:

Heuristic	Rule Specification	Impact on User
Deload Gate	Every 20 completed sessions (`total % 20 == 0`), the engine forces 3–5 days of `MOBILITY` routines.	Prevents chronic systemic fatigue and tendon strain; promotes active recovery.
Overlap-Aware Recovery	Treats `FULL_BODY` as overlapping Lower, Upper, and Core. Back-to-back identical muscle stress applies a **-15 penalty**; alternating groups grants a **+12 bonus**.	Prevents consecutive-day quadriceps or pushing overload.
Weekly Volume Guard	Penalizes any muscle focus area appearing $\ge 2\times$ (or Full Body $\ge 3\times$) in the rolling 7 completed sessions with a **-20 penalty**.	Enforces natural muscle rotation throughout the week.
Staleness Guard	Penalizes the exact workout completed in the prior session with **-8 penalty** to break score ties.	Guarantees routine variety even when user preferences are narrow.
Auto-Progression	≥ 3 of last 5 sessions rated `TOO_EASY` with $\ge 95\%$ completion advances level to Intermediate/Advanced. Also triggers if 3 consecutive sessions hit 100% completion.	Applies progressive overload without manual user configuration.
Auto-Regression	≥ 3 ratings of `TOO_HARD` or completion rate $< 60\%$ drops difficulty by one tier and activates low-impact exercise substitutions.	Protects struggling users from dropout and injury.

4.3 Real-Time Interval Workout Player

Powered by `WorkoutSessionEngine` operating on a dedicated coroutine ticker:

- **State Lifecycle:** `Idle` → `Countdown` (5s prep) → `ActiveExercise` (30–60s) → `RestInterval` (15–20s) → `Summary`.
- **Interactive Controls:** Play, Pause, Resume, Skip Next, Go to Previous Exercise.
- **Audio Cues:** Synthesized beeps via Android `ToneGenerator`: countdown tick (`TONE_PROP_BEEP`), exercise start (`TONE_PROP_ACK`), rest interval (`TONE_PROP_PROMPT`), halfway mark, and completion fanfare.
- **Haptic Engine:** Waveform vibration patterns coordinated by `WorkoutFeedbackCoordinator` for silent pocket/floor workout execution.
- **Visual Motion Guides:** Animated demonstration component and live countdown circular progress rings.

4.4 Exercise Replacement & Regression Engine

Enables users to swap out problematic exercises prior to or during a workout. Substitutions preserve physiological intent while adjusting joint loading:

- *Jumping Jacks (Moderate Impact)* → **Arm Circles Warm-up** (Zero Impact)
- *Mountain Climbers (Moderate Impact)* → **Forearm Plank Hold** (Low Impact Core)

- *High Knees (High Impact)* → **Glute Bridges** (Low Impact Posterior Chain)
- *Standard Push-ups* → **Wide-Grip Push-ups** or **Knee Push-up regression**
- *Forward Lunges (Shear force on knees)* → **Reverse Lunges** or **Wall Sit**

4.5 Content Catalog: Exercises, Workouts & Programs

Fit10 launches with a curated catalog of **21 bodyweight exercises**, **10 structured 10-minute workouts**, and **2 multi-week programs**:

Exercise	Category	Impact	Target Musculature	Default Cadence
Bodyweight Squats	STRENGTH	Low	Quadriceps, Glutes, Hamstrings	15 reps / 45s
Alternating Forward Lunges	STRENGTH	Low	Quadriceps, Glutes, Calves	14 reps / 45s
Alternating Reverse Lunges	STRENGTH	Low	Glutes, Hamstrings (Knee-friendly)	14 reps / 45s
Glute Bridges	STRENGTH	Low	Glutes, Hamstrings, Erector Spinae	15 reps / 45s
Glute Bridge March	STRENGTH	Low	Glutes, Anti-rotation Core	12 reps / 45s
Wall Sit	STRENGTH	Low	Isometric Quadriceps, Glutes	45s hold
Push-ups	STRENGTH	Low	Chest, Triceps, Anterior Deltoids	12 reps / 45s
Wide-Grip Push-ups	STRENGTH	Low	Outer Pectorals, Deltoids	10 reps / 45s
Diamond Push-ups	STRENGTH	Low	Triceps, Inner Chest	8 reps / 45s
Prone Y-T-W Hold	STRENGTH	Low	Rear Delts, Rhomboids, Trapezius	45s hold
Forearm Plank	CORE	Low	Transverse Abdominis, Deep Core	45s hold
Bicycle Crunches	CORE	Low	Internal/External Obliques	20 reps / 40s
Plank Shoulder Taps	CORE	Low	Shoulder Stability, Anti-rotational Core	45s
Bird Dog	CORE	Low	Rotary Stability, Lower Back	45s
Superman Hold	CORE	Low	Erector Spinae, Glutes, Posterior Chain	45s hold
Jumping Jacks	CARDIO	Moderate	Cardiovascular, Calves, Deltoids	40s
Mountain Climbers	CARDIO	Moderate	Hip Flexors, Core, Shoulder Girdle	40s
High Knees	CARDIO	High	Hip Flexors, Fast-twitch Calves	40s
Arm Circles Warm-up	WARMUP	Low	Rotator Cuff, Deltoids	30s
Cat-Cow Stretch	MOBILITY	Low	Spinal Flexion/Extension, Neck	45s flow

Exercise	Category	Impact	Target Musculature	Default Cadence
Child's Pose	COOLDOWN	Low	Latissimus Dorsi, Hips, Spinal Decompression	45s hold

Workouts & Program Tracks

- **Week 1 Foundation Workouts:** Full Body Ignition, Core Shred, Lower Body Burn, Upper Body Sculpt, Daily Mobility Reset.
- **Week 2 Overload Workouts:** Full Body Power (+2 reps / +5s holds), Core Endurance, Lower Body Burn Level Up, Upper Body Strength, Full Body Challenge (HIIT finisher).
- **14-Day Fit10 Kickstart:** 2-week beginner progression with built-in active recovery days on Days 4, 7, and 9.
- **14-Day Fit10 Intermediate:** Dense volume progression for users graduating from Kickstart.

4.6 Monetization & Subscription Strategy (Fit10 Pro)

Built on **Google Play Billing Library v9.1.0** conforming to Play Store billing policies:

Tier	Price (INR)	Trial / Billing Terms	Access Level
Annual Pass (Recommended)	₹499 / year	**7-Day Free Trial**, auto-bills ₹41.50/mo equivalent.	Full Pro Access to all 15+ tracks, AI adaptation engine, exercise swaps, ad-free.
Monthly Pass	₹149 / month	Billed monthly with instant cancellation.	Full Pro Access on a flexible month-to-month basis.
Free Tier (Ad-Supported)	₹0	No credit card required.	Access to foundation week workouts, standard player, and manual logging.

5 Technical Architecture & Data Design

> **Architecture Philosophy: Clean MVVM + Unidirectional Data Flow (UDF)**
>
> Every feature is decomposed into **UI (Compose)**, **ViewModel (StateFlow)**, **Domain (Use Cases / Engines)**, and **Data (Room / DataStore)**. UI components observe immutable UI states and dispatch declarative events.

5.1 Database Entity Relationship (Room)

- `ExerciseEntity`: Stores 21 canonical exercises (ID, name, instructions, tips, target muscles, equipment, impact level).
- `WorkoutEntity & WorkoutExerciseEntity`: Represents workout metadata and joined exercise sequence with order index, duration, reps, and rest timers.
- `ProgramEntity & ProgramDayEntity`: Multi-week tracks mapped to daily workout IDs and rest flags.
- `WorkoutResultEntity`: Historical session logs (session ID, workout ID, duration, exercises completed, calories, RPE feedback rating, completed timestamp).

6 Non-Functional Requirements (NFRs)

Dimension	Target Standard	Verification Method
Cold Start Latency	$\le 800\text{ ms}$ to interactive Home screen.	Baseline Profiles + R8 optimization.
UI Fluidity	Sustained 60 FPS across all workout countdowns and timer animations.	Compose Layout Inspector / Macrobenchmark.
Offline Resiliency	100% of workouts, timers, audio tones, and logs execute without internet.	Room database pre-seeded on first run.
Thermal & Battery Load	$< 2\%$ battery consumption per 10-minute session.	Coroutine timer delays over wake-locks; low CPU audio synthesis.
Data Privacy	Zero health/biometric data transmitted to remote servers by default.	On-device Room storage; exportable JSON database.

7 Key Performance Indicators (KPIs)

$\ge 82\%$	$\ge 45\%$ / 25%	$\ge 12\%$
WORKOUT COMPLETION	**D7 / D30 RETENTION**	**TRIAL CONVERSION**
Percentage of initiated workouts that finish without early abandonment.	Sustained habit retention enabled by 10-minute micro-commitment.	Conversion rate from the 7-day annual pass trial into paying subscribers.

8 Product Roadmap (Post-v1.0)

- **v1.1 (Wear OS Companion):** Standalone smartwatch workout player with real-time wrist haptics and heart rate monitoring.
- **v1.2 (Health Connect Integration):** Bidirectional sync with Google Health Connect for unified caloric and step tracking.
- **v1.3 (Chromecast / Smart TV Casting):** Full-screen animated workout timer view designed for living room workouts.
- **v1.4 (Custom Circuit Builder):** Power users can assemble custom 10-minute workout circuits from the 21 exercise catalog.